

1. Purpose

This document reports the dataset assembled in Phase 4 against the four hypotheses defined in the Research Plan (Deliverable 3), and states a verdict for each.

2. A note on two distinct "leakage" numbers in this dataset

This dataset contains two conceptually different leakage figures, and conflating them is a common error worth heading off explicitly:

- **Circuit-level static leakage** (DC/ICC2 reports below): the leakage of the synthesized/placed netlist as a static circuit, as if permanently powered. This reflects the *cost* of the controller's own logic — relevant to H4.
- **Runtime behavioral leakage savings** ([tb_comparison.v](#) simulation, Table C below): a cycle-weighted estimate of leakage *saved* by actually entering sleep states during operation, using the literature-derived weights (ON=1.0, LS=0.81, DS=0.64). This is the actual power-gating benefit claim — relevant to H1, H2, H3.

A design can simultaneously have higher static leakage (more controller logic) and still deliver real runtime savings (because it sleeps effectively) — that is in fact exactly what this dataset shows, and it is not a contradiction.

3. Table A — DC Synthesis PPA (pre-layout estimate)

	Baseline (static_pg_top)	Proposed (adaptive_pg_top)	Overhead
Area, LP corner (5 ns)	350.98 μm^2	422.51 μm^2	+20.4%
Area, HS corner (2 ns)	351.25 μm^2	442.76 μm^2	+26.1%
Timing, both corners	MET	MET	—

Dynamic power figures are available in [reports/{baseline,proposed}/power_*.rpt](#) for the full paper-stage PPA table; omitted here as not required by any of H1–H4.

4. Table B — ICC2 Post-Route PPA (real, LP corner)

	Baseline	Proposed	Overhead
Post-route area	361.50 μm^2	410.61 μm^2	+13.6%
Post-route instance count	1014	1118	+10.3%
Post-route timing (WNS / TNS)	0.0000 / 0.0000 (MET)	0.0000 / 0.0000 (MET)	Equal
Post-route static leakage (ICC2 QoR, pW)	115164.17	138427.83	+20.2%

The post-route area overhead (13.6%) is materially lower than the pre-layout DC estimate (20.4–26.1%) — DC's pre-layout area includes a wireload-model interconnect estimate that overstates real cost relative to what the router actually achieves. **Table B, not Table A, is the authoritative H4 number.**

5. Table C — Functional Comparison ([tb_comparison.v](#))

Metric	BURST — Base / Prop	SPARSE — Base / Prop	MIXED — Base / Prop
Cycles ON	78 / 135	8 / 16	92 / 104
Cycles LS	57 / 0	10 / 11	10 / 9
Cycles DS	0 / 0	155 / 146	81 / 70
Leakage saving	8.8% / 0.7%	33.4% / 31.6%	17.0% / 14.7%
Mode transitions	19 / 0	26 / 23	20 / 17
Wake-up events	9 / 0	8 / 7	7 / 6
Throughput	19.3% / 59.3%	0.0% / 0.6%	38.3% / 38.8%
Correctness (mismatches)	0	0	0

6. Hypothesis Verdicts

H1 — CONFIRMED. Under BURST, the baseline thrashes (19 transitions, 9 wake-ups, throughput collapses to 19.3%) while the proposed design's activity-rate gate suppresses sleep entirely (0 transitions, throughput preserved at 59.3%). This is the headline result of the project.

H2 — CONFIRMED. Under SPARSE, leakage savings converge (33.4% vs 31.6%, within 1.8 points) — the proposed design is not overly cautious when idle is genuinely unambiguous.

H3 — Directionally supported, primary evidence from BURST. Energy proxy (transitions $\times$ max wake-up latency, as a wake-up-cost unit): under BURST, baseline accrues 19 transitions $\times$ 3-cycle max latency = 57 wake-up-cycle-equivalents of overhead to achieve its 8.8% leakage saving; proposed accrues $0 \times 0 = 0$ wake-up-cycle-equivalents for its 0.7% saving. Baseline's leakage advantage (8.1 percentage points) is purchased entirely at the cost of 57 wake-up-cycle-equivalents and a throughput collapse to a third — this is the clean case for H3. On MIXED, baseline edges proposed on raw leakage saving (17.0% vs 14.7%) but at higher transition cost (20 vs 17); this trade-off is real but smaller and not unambiguous on its own. **Recommendation for the paper: lead with BURST as the primary H3 evidence, present MIXED as a secondary, more nuanced data point.**

H4 — REFUTED, narrowly. Real post-route area overhead is 13.6% (Table B), exceeding the <10% target. Timing closure is identical and perfect on both designs (WNS=TNS=0), so the timing half of H4's falsification criterion is not triggered — only the area half is. This refutation is mild relative to the pre-layout estimate (20–26%) and should be framed in the paper as a bounded, modest cost in exchange for the H1 result, not as a failure of the "lightweight" design goal outright — see the reframing discussed earlier: *comparator-based adaptive control with bounded overhead*, not *adaptive control at near-zero cost*.

7. Limitations

1. Static/post-route leakage numbers (Tables A, B) reflect the controller's circuit-level cost, not runtime gating behavior — see Section 2. Separately, Table C's leakage-saving percentages use literature-derived per-mode weights (Agarwal et al., ON=1.0/LS=0.81/DS=0.64) applied identically to both designs, not this project's own measured per-domain leakage. This keeps the *comparison* between baseline and proposed valid (same weights, same method, both sides), but absolute savings percentages should be read as comparative, not as grounded absolute energy claims.

2. ICC2 post-route data exists only for the LP corner; HS corner P&R was not run (DC-level HS data only).
3. Corner values (5 ns/2 ns) deviate from the originally locked Research Plan (15 ns/8 ns) — see Deliverable 4 §5 for justification.
4. The H3 "total energy" metric is a derived proxy (leakage-saving % combined with transition-cost), not a first-principles joule measurement — appropriate for this project's scope but worth stating plainly.

8. Conclusion

Three of four hypotheses are confirmed or directionally supported with real data; H4 is refuted but narrowly, with a defensible reframing available for the paper. The dataset is sufficient to proceed to Deliverable 6 (IEEE Paper).